

Jason Lin

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EDUCATION

University of Toronto

Sept. 2022 — Expected Jun. 2027

B.Asc. in Computer Engineering + PEY, minor in Artificial Intelligence Engineering

- **Relevant Courses:** Fund. to Deep Learning and Artificial Intelligence, Operating Systems, Algorithms & Data Structures
- **Extra-Curricular Involvements:** UofT Badminton Club (Club President), Association of Chinese Engineers (Department Lead)

TECHNICAL SKILLS

Languages

Python, SQL, C++, C, Verilog, SystemVerilog, Java

Tools and Frameworks

Pandas, Anaconda, Docker, Streamlit, Git, GitHub, PyTorch, TensorFlow, Jupyter, MS SQL Server

EXPERIENCE

Bristol Gate Capital Partners Inc.

May 2025 — May 2026

Software Engineering Co-op

Toronto, ON

- Refactored a legacy financial ETL pipeline (**10+ components**) to enable scalable processing of large-scale market data
- Designed and deployed a centralized **Streamlit-based** data platform, consolidating scattered data sources and process modules, reducing manual data handling overhead by **60+%**
- Built a scalable **backtesting** framework for financial models, enabling **end-to-end** validation across US and CA market datasets
- Designed and executed large-scale backtesting experiments across **25+** years of US and CA market data, quantifying the impact of model inputs on stock selection outputs to improve model performance and data integrity

University of Toronto, FORCOLAB

May 2024 — May 2025

Undergraduate Research Intern, Supervised by Prof. Shurui Zhou and David Lie

Toronto, ON

- Investigated and proposed potential solutions for an existing **open-source cybersecurity vulnerability** (CVE-2024-3094)
- Curated a dataset consisting of 2000+ user conversations from GitHub and XZ mailing list using **GitHub API** and **Requests**
- Utilized inferences with **Hugging Face Models** to quantify sentimental behavior for more than **200 GitHub PRs and issues**.
- Leveraged **Regression Discontinuity Design (RDD)** to quantify the impact of 100 user' interventions across 10 GitHub repository.
- Constructed an interactive tool with **PyVis** to visualize user's sentiment relationships to extract user-specific attributes.

Zhuhai Pilot Technology Co., Ltd.

July 2023 — August 2023

Machine Learning Engineering Intern

Zhuhai, Guangdong

- Developed a prediction algorithm in **Python** and **Scikit-Learn** to predict required cooling load in building complexes
- Iterated with various ML algorithms such as **Linear Regression**, **XGBoost**, and **LightGBM**, achieving **97% accuracy** in prediction
- Leveraged feature Engineering to add statistical features to the dataset, which enhanced accuracy by **8%**
- Trained a separate data mining model using **Random Forest** for data filling, which further improved prediction accuracy by **12%**

PROJECTS

Moodify | Demo

May. 2024 — Sept. 2024

- Architected a recommendation model that recommends songs based song features, lyrical sentiment and the user's mood
- Created a lyrics fetcher with **Genius API** to collect and pre-process lyrical data for over **10000 songs**
- Trained and optimized hyperparameters for an **RNN** sentiment analysis model using **PyTorch** with a **90%+ sentimental accuracy**
- Coordinated, integrated, and deployed the model to users, achieving a **96% user satisfaction rate**

StudentSphere | Demo

Jan. 2024 — Apr. 2024

- Lead and developed city mapping **GIS** software in **C++** utilizing the **STL Library** to process and manage over **20 Million Data Points** from **Open Street Map** database
- Implemented pathfinding algorithms such as **Dijkstra's** and **A***, achieving **top 10% performance** for solving the Traveling Salesman Problem
- Designed and developed the front-end UI/UX of the mapping system using **EZGL**, resulting in **94% positive user feedback**
- Optimized backend display algorithms and data structures, achieving a consistent **30+FPS** in real-world use cases.

AWARDS & CERTIFICATIONS

Edith Grace Buchan Undergraduate Summer Research Fellowship

May 2024 — May 2025

University of Toronto

- Faculty of Electrical and Computer Engineering Summer Research fellowship award, with a value of approximately **\$5,380**